SOFTWARE REQUIREMENTS SPECIFICATION

Web-Based ML Model Evaluator | Complete Requirements Table | v1.1 (Corrected)

Actors and Stakeholders

The following actors interact with the system and are defined in the UML Diagram and Use Case Flow document. All role-scoped requirements in this SRS use these actor definitions.

Actor	Role Description
Student	Primary user. Uploads datasets, configures and trains models, views and compares evaluation results, and downloads outputs.
Instructor	Limited reviewer. May view evaluation results explicitly made available for review. Cannot upload datasets, start training, or modify any stored data.
Admin	Administrator. Manages user accounts, dataset records, and system activity logs. Has access to administrative functions denied to Students and Instructors.
ML Service	Internal backend component. Executes model training and evaluation when triggered by an authenticated Student request. Not a human actor.

Note: *The system shall enforce role distinctions server-side. A user's assigned role (Student, Instructor, or Admin) shall be stored in the user account record and checked by the backend on every protected request. Frontend-only role hiding does not satisfy this requirement (see NFR-4).*

Functional and Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria (PASS/FAIL)	Rationale
FR-1.1	Functional	The system shall allow a new user to register using required account information.	High	PASS: A valid unique account is created. FAIL: Invalid input is accepted or the account is not created.	Enables access to the evaluator.
FR-1.2	Functional	The system shall authenticate registered users using valid login credentials.	High	PASS: Valid credentials establish a session and invalid credentials are rejected. FAIL: Authentication behaves otherwise.	Protects user accounts.
FR-1.3	Functional	The system shall allow an authenticated user to log out and terminate the active session.	Medium	PASS: The previous session cannot access protected resources after logout. FAIL: The session remains usable.	Prevents continued access after logout.
FR-1.4	Functional	The system shall restrict protected resources and operations to authenticated users with the required permissions.	High	PASS: Anonymous and unauthorized requests are denied. FAIL: Protected resources are accessible without permission.	Protects user-specific resources.
FR-2.1	Functional	The system shall allow an authenticated user to upload a dataset in CSV format.	High	PASS: A valid CSV is accepted and associated with its uploader. FAIL: Valid CSV is rejected or ownership is incorrect.	Provides input data for experiments.
FR-2.2	Functional	The system shall validate uploaded files for supported format, readability, and usable tabular structure.	High	PASS: Valid CSV is accepted; empty, malformed, or unsupported files are rejected with feedback. FAIL: Invalid input is accepted.	Prevents unusable input.
FR-2.3	Functional	The system shall display the selected dataset's column names and a sample of rows.	High	PASS: Preview matches the selected dataset. FAIL: Rows or column names are missing or incorrect.	Lets users inspect data.
FR-2.4	Functional	The system shall display basic dataset statistics, including row and column counts and available data-type and missing-value information.	Medium	PASS: Displayed statistics match the dataset. FAIL: Statistics are inaccurate or refer to another dataset.	Helps users understand data quality.

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FR-2.5	Functional	The system shall allow a user to replace or upload a dataset while preserving correct ownership and metadata associations.	Medium	PASS: Dataset metadata remains associated with the correct owner. FAIL: Ownership is lost or another user's data is exposed.	Supports dataset updates and integrity.
FR-3.1	Functional	The system shall allow users to configure and apply supported data-preprocessing operations before training.	High	PASS: Supported options are accepted and applied or recorded. FAIL: Invalid options are accepted or valid options fail unexpectedly.	Prepares data for ML.
FR-3.2	Functional	The system shall allow a user to select a valid target column from the dataset.	High	PASS: An existing usable column is saved as target. FAIL: A nonexistent or unusable column is accepted.	Defines the prediction target.
FR-3.3	Functional	The system shall allow a user to configure a supported train-test split.	High	PASS: A valid split is accepted and yields usable partitions. FAIL: An invalid split is accepted.	Separates training and evaluation data.
FR-4.1	Functional	The system shall present supported ML algorithms and allow selection of one compatible with the configured task and data.	High	PASS: A supported compatible algorithm is selected. FAIL: Unsupported or incompatible algorithm is accepted.	Enables algorithm selection.
FR-4.2	Functional	The system shall train the selected model using the chosen dataset and experiment configuration when training is started.	High	PASS: Training uses the selected configuration and reports status. FAIL: Incorrect configuration is used or status is absent.	Core evaluator function.
FR-4.3	Functional	The system shall record each training run's status and available configuration details and associate them with the correct user.	High	PASS: Run details are retrievable and correctly associated. FAIL: Details are missing, incorrect, or misassociated.	Supports traceability.
FR-5.1	Functional	The system shall calculate and display evaluation metrics appropriate to the completed run's task type.	High	PASS: Appropriate classification or regression metrics are shown. FAIL: Metrics are missing or inappropriate.	Measures model performance.
FR-5.2	Functional	The system shall display supported evaluation visualizations, such as a confusion matrix where applicable.	Medium	PASS: A compatible visualization reflects the selected run. FAIL: Visualization is incorrect or inapplicable.	Makes results easier to interpret.
FR-5.3	Functional	The system shall compare two or more accessible, compatible runs using comparable metrics and configuration details.	Medium	PASS: Comparable runs show correct metrics. FAIL: Incompatible runs are misleadingly compared.	Supports model selection.
FR-5.4	Functional	The system shall allow users to download supported model artifacts or evaluation reports for runs they may access.	Medium	PASS: The correct available output is downloaded. FAIL: Wrong or unauthorized output is returned.	Allows retention of results.
FR-6.1	Functional	The system shall display the authenticated user's available run history with recorded status and result details.	Medium	PASS: The user's runs and details are shown accurately. FAIL: Runs are missing, inaccurate, or expose another user's history.	Enables review of prior experiments.
FR-6.2	Functional	The system shall expose documented backend functionality through a defined API/interface for frontend requests.	Medium	PASS: Valid requests return the expected response structure and invalid requests return controlled errors. FAIL: Requests are handled inconsistently.	Defines frontend-backend communication.
FR-6.3	Functional	The system shall provide understandable error feedback for invalid input and failed operations without exposing sensitive internal details.	High	PASS: Actionable feedback is shown without secrets or stack traces. FAIL: Errors are silent or reveal sensitive details.	Improves usability and safe failure handling.
FR-7.1	Functional	The system shall provide authorized Admin users with the user-management functions included in the project scope.	Medium	PASS: Admin can access supported functions and other roles are denied. FAIL: Unauthorized access is allowed.	Supports administration.

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FR-7.2	Functional	The system shall provide authorized Admin users with access to dataset records and metadata for supported management.	Medium	PASS: Records are accurate and ownership associations remain intact. FAIL: Unauthorized access or inconsistent ownership occurs.	Supports dataset administration.
FR-7.3	Functional	The system shall provide authorized Admin users with read-only access to available application activity logs.	Low	PASS: Admin can view logs and non-Admins are denied. FAIL: Unauthorized access or application-level log editing is allowed.	Supports accountability.
FR-7.4	Functional	The system shall allow an Instructor to view evaluation results explicitly made available for review, without modifying datasets or runs.	Low	PASS: Instructor can view authorized results in read-only mode and cannot create, update, or delete any resource. FAIL: Unauthorized results are visible or write operations succeed for an Instructor.	Supports limited review.
FR-7.5 ★	Functional	The system shall assign each registered user a role (Student, Instructor, or Admin) and enforce role-based access control server-side on every protected request.	High	PASS: Each role can only access resources and operations defined for that role; a lower-privileged role cannot access a higher-privileged endpoint by calling the API directly. FAIL: Role enforcement is absent or bypassable.	Required to support the actor model defined in the UML and Use Case Flows. Without a role field, FR-7.1 through FR-7.4 cannot be enforced. ★ Added in v1.1.
NFR-1	Nonfunctional (Performance)	For supported datasets within the configured size limit, common UI actions and metadata/preview requests shall return within an agreed target under normal load; the target shall be measured and documented during testing.	High	PASS: Response-time measurements are recorded and compared with the agreed project target under normal load. FAIL: The agreed target is not met or no measurements are recorded.	Keeps common user actions responsive.
NFR-2	Nonfunctional (Performance)	The system shall provide visible progress or a clear running status for model-training operations that take longer than a normal request.	Medium	PASS: The user can identify when training is running, completed, or failed. FAIL: A long operation appears unresponsive without status feedback.	Improves transparency during training.
NFR-3	Nonfunctional (Security)	Passwords shall not be stored in plaintext; authentication credentials shall be handled using an appropriate secure password-storage mechanism.	High	PASS: Stored passwords use an appropriate secure representation and are not exposed in responses or logs. FAIL: Plaintext passwords are stored or exposed.	Protects user accounts.
NFR-4	Nonfunctional (Security)	Server-side authorization shall protect user-specific datasets, model runs, and downloadable artifacts; hiding a UI control alone shall not count as access control.	High	PASS: Unauthorized direct backend requests are denied. FAIL: Protected resources can be accessed by bypassing UI restrictions.	Prevents unauthorized access.
NFR-5	Nonfunctional (Privacy)	The system shall isolate each user's uploaded datasets, run history, metrics, and artifacts from other users unless explicit sharing is implemented.	High	PASS: Cross-user requests for restricted resources are denied. FAIL: Another user's datasets, runs, metrics, or artifacts are exposed.	Protects user data and results.
NFR-6	Nonfunctional (Reliability)	Invalid input or a failed training/export operation shall not corrupt previously saved datasets or completed run records.	High	PASS: Failed operations are reported and existing datasets/completed runs remain intact. FAIL: Previously saved data or runs are corrupted.	Maintains data integrity during failures.
NFR-7	Nonfunctional (Reproducibility)	The system shall retain sufficient run configuration, including algorithm and relevant split/preprocessing settings, to explain how a result was produced.	Medium	PASS: A prior run's stored configuration and results can be retrieved and interpreted. FAIL: Results cannot be linked to the configuration used.	Supports interpretation of previous experiment s.
NFR-8	Nonfunctional (Usability)	The primary workflow shall be navigable through a desktop web UI with clear labels, validation messages, and status feedback.	Medium	PASS: A tester can complete the primary workflow and understand validation/status feedback. FAIL: Core steps are unclear or feedback is missing.	Makes the application usable for its intended workflow.

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NFR-9	Nonfunctional (Compatibility)	The application shall support the agreed current versions of major desktop browsers; the exact test matrix shall be recorded for the release.	Medium	PASS: Core workflows operate in each browser/version in the agreed test matrix. FAIL: A core workflow is unusable in a supported browser.	Supports browser-based access.
NFR-10	Nonfunctional (Maintainability)	The implementation shall separate UI, API/business logic, data persistence, and model-training concerns into maintainable modules.	Medium	PASS: The major responsibilities are separated into identifiable modules. FAIL: Routine changes require modifying unrelated parts of the application.	Supports maintainable development.
NFR-11	Nonfunctional (Scalability)	The system shall enforce configurable upload/resource limits and handle datasets within the documented project capacity without uncontrolled resource use.	Medium	PASS: Configured limits are enforced and over-limit inputs receive controlled feedback. FAIL: Limits are ignored or resource use becomes uncontrolled.	Helps control resource consumption.
NFR-12	Nonfunctional (Data Integrity)	Dataset metadata, run configuration, metrics, and artifact references shall remain consistently associated with the correct user and run.	High	PASS: Metadata, results, and artifacts refer to the correct user and run. FAIL: References are missing, inconsistent, or misassociated.	Maintains trustworthy experiment records.
NFR-13	Nonfunctional (Error Handling)	Errors shown to users shall be understandable and shall not expose passwords, secret configuration, stack traces, or other sensitive internal details.	High	PASS: Error messages are actionable and omit sensitive internal information. FAIL: Errors expose secrets, stack traces, or other sensitive details.	Supports safe troubleshooting.
NFR-14	Nonfunctional (Portability)	The application shall be deployable in the documented project environment using its setup instructions and declared dependencies.	Medium	PASS: A clean environment can be set up using the documented instructions and declared dependencies, or gaps are recorded. FAIL: Setup instructions or dependency declarations prevent reproducible setup.	Supports reliable setup and deployment.

★ *FR-7.5 is new in v1.1. It is the foundational requirement that enables enforcement of FR-7.1, FR-7.2, FR-7.3, and FR-7.4.*